Opt on manifold Notes

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Chapter 1

Optimization theory on linear spaces

These are brief notes based on chapters 5 of [6] and chapter 4 and 5 of [4]. We consider optimization problems for functions defined on subsets of linear spaces. Some proofs are presented in a geometric manner. The section on duality refers to chapter 4 of [3] to illustrate an interesting relation between Fenchel duality and Lagrangian duality. For simplicity, we only consider proper functions: those taking values in $(-\infty, \infty]$ and not identically infinity.

1.1 Existence and Uniqueness

It's well-known that if f is lower semicontinuous and domain is compact, then f has a minimum. Thus if a closed sublevel set $C = \{f \leq c\}$ is bounded and nonempty for some $c \in \mathbb{R}$, then f has a minimum over C hence has a global minimum. A typical example is when f is coercive, i.e. $\lim_{|x|\to\infty} f(x) = \infty$, then the sublevel sets are bounded and nonempty for sufficiently large c.

On the other hand, if f is lower semicontinuous and strictly quasiconvex(for any segment connecting points of graph of f, the graph lies strictly below maximum of the endpoints; this implies the sublevel sets are strictly convex), then such minimum must be unique.

1.2 Optimization condition of unconstrained optimization

The smooth case is well-known: if f is differentiable, then a necessary condition for x to be a local minimum point is $\nabla f(x) = 0$ and $\mathrm{Hess} f(x) \succeq 0$ if f is twice-differentiable, and a sufficient condition is $\nabla f(x) = 0$ and $\mathrm{Hess} f(x)$ is positive-definite. These conditions also applied to functions on manifolds.

If the function is convex, then the necessary and sufficient condition for x to be a global minimum point is $0 \in \partial f(x)$. This is necessary even if f is not convex, with the definition of subdifferential suitably generalized.

1.3 Duality theory

From now on we consider the optimization problem with constraints, the duality theory is very useful to understand the optimality condition of such problems. The idea is to convert the problem to dual problem with better constraints and computable optimization condition.

1.3.1 Fenchel duality

So basically we are extending Legendre transform to general cases: the Frenchel dual of f is defined as

$$f^*(y) = \sup_x \langle x, y\rangle - f(x)$$

It can be understood as the restriction of the support function of the epigraph of f:

$$f^*(y) = \sup_{(x,t)\in \mathrm{epi}(f)} \langle (x,t), (y,-1)\rangle$$

which detects the shape of epigraph f by its supporting hyperplanes (although this only record information of its convex hull). This dual function is always convex and lower-semicontinuous as they are of a supremum of affine functions, and we have the Fenchel inequality:

$$f(x) + f^*(y) \geqslant \langle x, y\rangle$$

which immediately implies that the bidual $f^{**} \leq f$. If f is convex and lower-semicontinuous, then we have $f^{**} = f$. This is easy if we assume subdifferential exists, in which case we have nice summetry for points (x, y) that equalizing the Frenchel inequality: $x \in \partial f^*(y), y \in \partial f(x)$. Also, $*$ is order reversing, so in general, f^{**} is the largest lower-semicontinuous convex function that is less than or equal to f, i.e. epigraph of f^{**} is the convex hull of epigraph of f.

When function is smooth, f^* is just extension of Legendre transform. For example if $f(x) = e^x$ then

$$f^*(y) = \sup_x xy - e^x = \begin{cases} y\log y - y & y > 0 \\ 0 & y = 0 \\ \infty & y < 0 \end{cases}$$

An important non-smooth example is the indicator function on some set C:

$$I_C = \begin{cases} 0 & x \in C \\ \infty & x \notin C \end{cases}$$

we have

$$I_C{}^*(y) = \sup_{x\in C}\langle x, y\rangle - I_C(x) = \sup_{x\in C}\langle x, y\rangle$$

which is the support function of C. For a convex cone C, the dual of the support function is again a support function:

$$I_C^* = I_{C^0}$$

where

$$C^0 = \{y : \langle x, y\rangle \leq 0 \text{ for all } x \in C\}$$

is the polar cone of C.

1.3.2 Perturbation

This section refers to chapter VI of [5]. We consider a perturbation $F(x, \lambda)$ of f such that $F(x, 0) = f(x)$ and a family of optimization problem $p(\lambda) = \inf_x F(x, \lambda)$, then

$$p^*(\lambda) = \sup_s \left(\langle \lambda, s\rangle - \inf_y F(y, s) \right) = \sup_{s,y} \left(\langle \lambda, s\rangle - F(y, s) \right) = F^*(0, \lambda)$$

The primal program is $p(0) = \inf_x F(x, 0)$, the dual program is

$$p^{**}(0) = \sup_\lambda \left(\langle 0, \lambda\rangle - u^*(\lambda) \right) = \sup_\lambda \left(-F^*(0, \lambda) \right) = -\inf_\lambda F^*(0, \lambda)$$

This can be viewed as the promgram associated with F^*. If the bidual $F^{**} = F$, then $(P^{**}) = (P)$.

Henceforth, we omit the subscripts of inf and sup, understanding that $\inf F(\lambda) := \inf_x F(x, \lambda)$ and $\inf F^*(x) := \inf_\lambda F(x, \lambda)$. For example we shall write $p = \inf F$ and $p^{**} = -\inf F^*$. We have weak duality

$$\sup \left(-F^* \right) (0) \leq \inf F(0)$$

since $p^{**}(0) \leq p(0)$ and

$$p(0) - p^{**}(0) = \sup \left(-F^* \right) - \inf F$$

is the duality gap. If the gap is 0(this happens if p is convex and lower semicontinuous at 0), we say strong duality holds.

If F is convex, then $p(\lambda) = \inf_{x \in C} F(x, \lambda)$ is always convex as long as $p > -\infty$. This is because the epigraph of p is the projection of the epigraph of F, and the projection of a convex set is always convex.

The relative interior $\mathbf{Relint}(D)$ of a set D is the interior of D relative to its affine hull. Convex functions are continuous on the relative interior of their domain, so if $0 \in \mathbf{Relint}\,(\mathrm{dom}(p))$, then p is continuous at 0. In this case, the strong duality holds since p is continuous near 0. In this case, we say the convex program is strongly consistent, so basically we have a nice perturbation of f.

1.3.3 Lagrangian duality

It is heuristic to consider the smooth case first. Suppose we have a constrained optimization problem

$$\min_{x \in C} f(x) \text{ s.t. } g(x) = 0$$

If g has constant rank, then the first order condition is that df annihilates the tangent space $\mathrm{Ker} dg$, this happens iff $df = \langle \lambda, dg\rangle$ dualizing via a metric, we see

$$\nabla f(x) = \lambda \nabla g(x)$$

It has exactly the same first order optimization condition as the unconstrained minimizing problem of Lagrangian function

$$L(x, \lambda) = f(x) + \lambda g(x)$$

A constrained optimization problem is

$$\inf_x f(x) \text{ s.t. } g(x) \geq_K 0 \tag{P}$$

where K is a convex cone, for example a hemispace $H = [-\infty, 0]^p \times \{0\}^q$ We can convert the problem to an unconstrained optimization for

$$F(x) = f(x) + I_K(g(x))$$

where I_K is the indicator function on K.

Let's apply the insight of the last section to F. We consider the perturbation

$$F(x, \lambda) = f(x) + I_K(g(x) + \lambda)$$

Then

$$\begin{aligned}
-F^*(0, \lambda) &= \inf_{y,s} \left(-\langle \lambda, s\rangle + f(y) + I_K(g(y) + s)\right) \\
&= \inf_{y,r} \left(-\langle \lambda, r - g(y)\rangle + f(y) + I_K(r)\right) \\
&= \inf_{y} \left(f(y) + \langle \lambda, g(y)\rangle\right) - I_K^*(\lambda) \\
&= \inf_{y} \left(f(y) + \langle \lambda, g(y)\rangle\right) - I_{K^0}(\lambda)
\end{aligned}$$

Consider the Lagrangian:

$$L(y, \lambda) = f(y) + \langle \lambda, g(y)\rangle$$

with $\lambda \in K^0$ in mind. It can be under stood as an approximation of the original function $F(x) = f(x) + I_K(g(x))$ by a family of linear functions. We call

$$L(\lambda) = \inf_y L(y, \lambda)$$

the Lagrangian dual function.

Now we can write

$$-p^*(\lambda) = -F^*(0, \lambda) = \inf_y L(y, \lambda) - I_{K^0}(\lambda) = L(y) - I_{K^0}(\lambda)$$

The dual program of P is to minimize p^*, it can be written as

$$\sup_\lambda (L(\lambda)) \text{ s.t. } \lambda \leq_{K^*} 0 \tag{P^*}$$

In the hemispace case, the polar cone and dual cone is

$$H^0 = [0, \infty]^p \times \bar{\mathbb{R}}^q, H^* = -H^0 = [-\infty, 0]^p \times \bar{\mathbb{R}}^q$$

One can also interpret the Lagrangian dual from the perspective of minimax theory:

$$\sup_{\lambda\in K^0} L(\lambda) = \sup_{\lambda}\inf_{y}\left(L(y,\lambda) - I_{K^0}(\lambda)\right) \le \inf_{y}\sup_{\lambda}\left(L(y,\lambda) - I_{K^0}(\lambda)\right) = \inf_{g(y)\in K} f(y)$$

with equality at a saddle point of $L(y,\lambda), \lambda \in K^0$, in this case strong duality holds.

We shall see later that for convex problems, a saddle point will always be an optimal vector, the converse also true if Slater's condition holds.

There is a more genral duality theorem[5]: one have

$$\inf(f-g) = \sup\left(g_* - f^*\right)$$

when $f, g, f-g$ are proper convex. TBA

1.4 Optimization condition for general constraints

1.4.1 Tangent cone and constraint qualifications

The tangent cone $\mathscr{T}(x)$ (in the sense of the paratingent cone or contingent cone of Bouligand) at a feasible point x consists of all directions d such that there is a feasible sequence $x_k \to x$ and positive numbers t_k s.t.

$$(x_k - x)/t_k \to d$$

This should be equivalent to

$$\mathscr{T}(x) := \left\{d : \liminf_{t\to 0+} \frac{d(x+td, C)}{t} = 0\right\}$$

If the feasible set is convex, $\mathscr{T}(x)$ is the closure of the cone consists of open rays that intersect nontrivially the feasible set.

Cones of feasible directions defined above are not that computable. We need to introduce following linear approximations of $\mathscr{T}(x)$. Geometrically, these are

$$\mathfrak{T}(x) = (dF)^{-1}(\mathscr{T}_{g(x)}K)$$

where g is the constraint function and $\mathscr{T}_{g(x)}K$ is the tangent cone of the constraint set at $g(x)$. In general, we have $\mathscr{T}(x) \subset \mathfrak{T}(x)$, and the equality holds if the constraint is regular at x.

For simplicity, we restricted ourself to the case when the cone K is the simpliest manifold with corner H from now on, i.e. we are considering the problem

$$\inf_x f(x) \text{ s.t. } g(x) \le 0, h(x) = 0$$

At a fixed feasible point x, we rearrange the constraints $g = (g_1, g_2)$ s.t. $g_1(x) = 0, g_2(x) < 0$, so we say g_1, h are active constraints, and g_2 is inactive in a neighborhood of x. The necessary condition for a feasible direction is

$$\mathfrak{T}(x) = \{d : \langle \nabla h, d\rangle = 0, \langle \nabla g_1, d\rangle \le 0\}$$

This might be bigger than $\mathscr{T}(x)$, for example the linearized cone of feasible directions of the constraints $g_1(x) = x^3 \leq 0$ or $h(x) = x^3 = 0$ will give false feasible direction at 0. But if g_1, h is regular at x (namely its differential has full rank), then the level set $g_1 = 0, g_2 \leqslant 0$ defines a manifold with corner near x. In this case $\mathscr{T}(x) = \mathfrak{T}(x)$. We say Linear independence constraint qualification(**LICQ**) holds at x if the gradients of active constraints are linearly independent.

We actually do not need $(\nabla g_1, \nabla h)$ to have full rank, constant rank theorem tells us that it suffices the rank of each subset of the gradients of the active constraints at a vicinity of x is constant(**CRCQ**). In this case, for a fixed $d \in \mathfrak{T}(x)$, the active constraints with vanishing gradient along d will contains a submanifold that tangent to d, thus we can choose a curve along d that lies within such submanifold, while the rest of the active constraints with non-vanishing gradient along d will become inactive along such curve. A special case of these is linear constraints qualification(**LCQ**), when g_1, h are affine functions.

We also say Mangasarian-Fromovitz constraint qualification (**MFCQ**) holds at x if the gradients ∇h of equality constraints are linearly independent and there is a direction d_0 s.t. $\langle \nabla h, d_0 \rangle = 0, \langle \nabla g_1, d_0 \rangle < 0$ at x. In this case we choose curves $x(t, s)$ along $d + sd_0$ that lies within the submanifold defined by $h = 0$, in orther words

$$x(t, s) - x(0, 0) = (d + sd_0)(t + o(t))$$

now $x(t, t)$ will be a curve in the feasible set along d.

1.4.2 First order condition: Karush-Kubn-Tucker condition

At a local minima, either the function doesn't descent, or it meets the boundary of the feasible set. Thus if our linearized tangent cone is nice:

$$\mathscr{T}(x) = \mathfrak{T}(x)$$

then the following

$$\langle d, \nabla f \rangle < 0, \langle d, \nabla h \rangle = 0, \langle d, \nabla g_1 \rangle \leq 0$$

cannot be true at the same time at x.

Farkas lemma says these conditions are equivalent to the existence of λ and nonnegative μ s.t.

$$-\nabla f = \lambda \cdot \nabla h + \mu \cdot \nabla g_1$$

(here I use $\cdot$ to distinguish the product w.r.t. the index of constraints with the inner product on space) To see this, consider the convex cone

$$C = \{\lambda \cdot \nabla h + \mu \cdot \nabla g_1 : \mu \geq 0\}$$

the second and third conditions are equivalent to $-d \in C^*$, for any such d, $\langle d, \nabla f \rangle \geq 0$ thus $-\nabla f$ must lies in C^{**}. But the bidual of a closed convex cone is itself(by the hyperplane separation theorem), so $-\nabla f \in C$.

Now we understand that a necessary condition for a local minima x is the existence of λ and nonnegative μ s.t. satisfies $\partial_x L(x, \lambda, \mu) = 0$. We can put these conditions together with the information of active constraints: the **KKT condition** is

Stationarity:	$\partial_x L(x, \lambda, \mu) = \nabla f(x) + \lambda \cdot \nabla h(x) + \mu \cdot \nabla g(x) = 0$
Primal feasibility:	$g(x) \leq 0,\ h(x) = 0$
Dual feasibility:	$\mu \geq 0$
Complementary slackness:	$\mu \cdot g(x) = 0$

(λ, μ) are called the KKT multipliers. The KKT condition is necessary for local optimality if the linearized cone of feasible directions is nice, which amounts to verify various constraint qualifications.

The KKT condition is also natural in terms of the Lagrangian duality: it's the first order condition of the dual problem

$$\sup_{\lambda, \mu \in K^0} \inf_x L(x, \lambda, \mu)$$

Thus, KKT conditions are necessary if the strong duality holds.

1.4.3 Second order condition

The cone of critical directions is defined as

$$\mathscr{C}(x) = \{d \in \mathfrak{T}, \langle d, \nabla g_1 \rangle = 0\}$$

A second order necessary condition for local optimality at a KKT point (x, λ, μ) is that $\mathrm{Hess}_x L(x, \lambda, \mu)$ is positive semidefinite on $\mathscr{C}(x)$. A second order sufficient condition for local optimality is that $\mathrm{Hess} L(x, \lambda, \mu)$ is positive definite on $\mathscr{C}(x)$.

1.5 Optimization condition for convex problems

1.5.1 Slater's qualification and strong duality

Now we consider the problem P with convex objective function and affine equality and convex inequality constraints:

$$\min_{x \in D} f(x) \text{ s.t. } g(x) \leq 0, Ax = b$$

where f is proper and convex, g is convex and A is a linear map. As before, for convex problems we do not require the function to be differentiable. The perturbation of this problem is

$$F(x, \lambda, \mu) = f(x) + I_H((g(x), Ax - b) + (\lambda, \mu))$$

where $H = [-\infty, 0]^p \times \bar{\mathbb{R}}^q$ is the hemispace. The Lagrangian function is

$$L(x, \lambda, \mu) = f(x) + \lambda \cdot (Ax - b) + \mu \cdot g(x)$$

and

$$p(\lambda, \mu) = \inf_x \left(f(x) + I_H((g(x), Ax - b) + (\lambda, \mu))\right)$$

and dual function

$$p^*(\lambda, \mu) = -L(\lambda, \mu) + I_{H^0}(\lambda, \mu)$$

Recall our dicussion on perturbation and strong duality in 1.3.2. The condition $0 \in \mathbf{Relint}\,(\mathrm{dom}(p))$ is equivalent to the following condition: there is a point x s.t.

$$g_0(x) = 0, g_1(x) < 0, Ax = b$$

where g_0 is the affine part of g. This is the **Relaxed Slater's qualification** for the problem P. A special case is when $x \in \mathbf{Int}(\mathrm{dom}(p))$, this is equivalent to **Slater's qualification**

$$g(x) < 0, Ax = b$$

In these cases, the strong duality holds since p is continuous at 0.

1.5.2 Convex KKT

When Slater's condition holds, the strong duality holds, we have seen that if the functions are differentiable, the KKT condition is necessary for optimality. Note that MFCQ is a special case of this.

The convex case is slightly different. The KKT condition for convex problems is the same as before but with the differential replaced by subdifferential:

Stationarity:	$0 \in \partial_x L(x, \lambda, \mu) = \partial f(x) + \lambda \cdot A + \mu \cdot \partial g(x)$
Primal feasibility:	$g(x) \leq 0,\ Ax = b$
Dual feasibility:	$\mu \geq 0$
Complementary slackness:	$\mu \cdot g(x) = 0$

Convex KKT is sufficient for strong duality and optimazation. At the KKT point $L(x, \lambda, \mu) = f(x) \geqslant \inf f$, which imlies the duality gap is zero and the solution is optimal.

Under Slater's condition, the convex KKT condition is also necessary for optimality, but the proof should involve subdifferentials (TBA).

Chapter 2

Algorithms

2.1 First order algorithms

2.1.1 Gradient descent

2.1.2 Barzilai-Borwein

2.2 Second order algorithms

2.2.1 Newtonian

2.2.2 Trust-region method

2.3 Stochastic gradient descent

Chapter 3

First order optimization on manifolds

These are notes for chapter 4 and 11 of [1].

3.1 Riemannian gradient descent

Let M be a Riemanian manifold, and $f : M \to \mathbb{R}$ be a smooth function. Let R_x be the retraction at $x \in M$:i.e. $R_x : T_xM \to M$ is a smooth map s.t. $R_x(0) = x$ and $dR_x = id$. The Riemannian gradient descent (RGD) is defined as the iteration of the following update rule:

$$x_{k+1} = R_{x_k}(-t_k \nabla f(x_k))$$

where t_k is the step size.

3.1.1 Regularity and convergence

If f is bounded below, and if each step we have enough descent:

$$f(x_k) - f(x_{k+1}) \geqslant c|\nabla f(x_k)|^2 \tag{A}$$

Then

$$\min_{0 \leq k \leq N} |\nabla f(x_k)|^2 \leq \frac{f(x_0) - f(x_{N+1})}{c(N+1)} \leq \frac{f(x_0) - \inf f}{c(N+1)}$$

In particular the gradient will converge to zero, and any accumulation point of the sequence $\{x_k\}$ will be a critical point of f.

It's more natural to write the condition A as the Armijo-Goldstein condition

$$f(x) - f(R_x(-t\nabla f(x))) \geq rt|\nabla f(x)|^2 \tag{B}$$

where t is the step size, r is the ratio of the real descent and expected descent.

This can be done by control the "second order behavior" of f: if f is L-smooth w.r.t. the retraction, i.e.

$$f(R_x(s)) \leq f(x) + \langle \nabla f(x), s \rangle + \frac{L}{2}|s|^2 \tag{C}$$

for every $(x, s) \in TM$, then the descendence condition A holds if

$$t \in [t_{\min}, t_{\max}] \subset (0, 2/L)$$

In this case c can be chosen as

$$c = \inf \left\{ t - \frac{Lt^2}{2} \right\} = \min \left\{ t_{\max} - \frac{L}{2} t_{\max}^2, t_{\min} - \frac{L}{2} t_{\min}^2 \right\}$$

For example, if we choose $t_k = 1/L$, then $c = 1/(2L)$.

3.1.2 Backtrack line-search

In practice, we can not always find a fixed step size s.t. the condition A holds and we don't know a constant L beforehand. We can use backtrack line-search: we start with some initial step size t_0, and check if the condition B holds. If not, we shrink the step size by multiplying it with some $\tau \in (0, 1)$ until B is satisfied. The algorithm will terminate before

$$t \leq \frac{2(1-r)}{L}$$

after computing at most

$$\max \left(1, 2 + \log_\tau \left(\frac{2(1-r)}{tL} \right) \right)$$

retractions and evaluations as long as f satisfies the L-smoothness condition C.

Now we can choose some initial step size sequence t_k that $\sum_{k=0}^{\infty} t_k = \infty$, then at each step we have the descendence condition A with

$$c_k = r \min \left\{ t_k, \frac{2\tau(1-r)}{L} \right\}$$

As a consequence, we have

$$\min_{0 \leq k \leq K} |\nabla f(x_k)|^2 \leq \frac{f(x_0) - \inf f}{\sum_{k=0}^{K} c_k}$$

which implies $\lim_{k \to \infty} |\nabla f(x_k)| = 0$.

3.1.3 Local convergence

3.2 Convex optimization

In manifold case, we can define (strongly) convexity and Lipschitz smoothness in terms of geodesic segments: sets are convex if all geodesic segments connecting any two points in the set lie entirely

within the set, and functions are convex, strongly convex or Lipschitz smooth if their restriction to any geodesic is convex, strongly convex or Lipschitz smooth.

An alternative way to define μ-strong convexity is that $f(x) - f(y) - \mu\frac{d(x,y)^2}{2}$ being convex w.r.t. x. If x, y is joined by a minimizing geodesic with velocity $s \in T_yM$, let

$$D_f(x, y) = f(x) - f(y) - \langle \nabla f(y), s\rangle$$

we say f is μ-strongly convex if $D_f(x, y) \geq \frac{\mu}{2}d(x, y)^2$ where $d(\cdot, \cdot)$ is the Riemannian distance. This slightly different from the previous definition, but they are equivalent since the condition is essentially local: the function is (μ-strongly) convex if and only if it is (μ-strongly) convex at an open cover of S.

Bibliography

[1] Nicolas Boumal. An introduction to optimization on smooth manifolds. Cambridge University Press, 2023.

[2] Hu, J., Liu, X., Wen, ZW. et al. A Brief Introduction to Manifold Optimization. J. Oper. Res. Soc. China 8, 199-248 (2020).

[3] Dimitri P. Bertsekas. Convex Optimization Theory. Athena Scientific, 2009.

[4] Stephen Boyd and Lieven Vandenberghe. Convex Optimization. Cambridge University Press, 2004.

[5] Rockafellar, R. T. Convex Analysis. Princeton University Press, 1970.

[6] 袁亮、黄红选、王晓军. 最优化：建模、算法与理论. 高教出版社, 2022.